CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 1: Multithreaded Producer-Consumer Problem

Course: Advanced Java Programming Laboratory [23CS3504]
Category: Experiments / Practicals | Experiment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Implement thread synchronization, mutual exclusion, and inter-thread signaling using wait and notify .

2. Practical Tasks & Verification Steps:

1. Define shared circular buffer class with maximum capacity threshold of 5 elements.
2. Implement Producer thread writing integer values into buffer and calling wait when full.
3. Implement Consumer thread reading values from buffer and signaling notifyAll when space frees.
4. Execute multi-threaded test with 3 producers and 2 consumers verifying zero race conditions.

Submission Guidelines & Verification:

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